

NIPUN GAHANE

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EDUCATION

B.TECH – Computer Science Engineering, SRM Institute of Science and Technology Apr 2025

Specialization in **Software Engineering**

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Design, Database Management Systems, **Machine Learning**

GPA: 8.2 / 10.0

EXPERIENCE

Standard Chartered GBS - Bangalore

Technology - *Apprentice (Machine Learning & AI Focus)*

Jul 2025 – Present

- Contributing to the high-level design and architectural planning of an enterprise **Retrieval-Augmented Generation (RAG)** chatbot intended to automate internal banking queries.
- Collaborating with cross-functional teams to evaluate **Large Language Model (LLM)** capabilities, **vector database** solutions, and integration strategies for production deployment.
- Built and deployed a scalable **Spring Boot** API to facilitate high-throughput data remediation, integrating **JPA** data operations and automated **CI/CD** workflows via **Azure Pipelines**.

PROJECTS

Finance RAG App (SEC 10-K Q&A)

Mar 2026

Built an end-to-end financial **RAG** platform to ingest SEC filings, generating grounded answers with source citations via a **FastAPI** interface. Implemented hybrid retrieval using **ChromaDB** vector search and BM25 sparse search with RRF fusion, improving Recall@5 by 52% versus a vector-only baseline. **Tech: Python**, FastAPI, Streamlit, Ollama, LangChain, ChromaDB, Sentence-Transformers, PyTorch.

Efficient LLM Finetuning (Google Gemma 2B)

Oct 2025

Implemented parameter-efficient finetuning (**PEFT**) on the Google Gemma 2B model using **LoRA** and 4-bit **nf4** quantization. Trained the **LLM** on specialized datasets to adapt generative capabilities for specific stylistic outputs. **Tech: PyTorch**, Hugging Face (Transformers, TRL, PEFT), **LoRA**, Quantization.

End-to-End MLOps Pipeline

Feb 2024

Designed and automated an **MLOps** pipeline to manage the complete model lifecycle for predictive analytics, achieving an R-squared of 88.28% while ensuring continuous integration of data processing steps. **Tech: TensorFlow**, scikit-learn, Flask, MLflow.

Cloud-Native Backend & Data System

Mar 2026

Engineered a cloud-native backend using **Spring Boot** and **MongoDB Atlas**, deploying scalable microservices via **Docker** and **AWS ECS (Fargate)**. Integrated **GitHub Actions** CI/CD to ensure robust delivery for data applications. **Tech: Java**, Spring Boot, AWS, Docker, MongoDB.

TECHNICAL SKILLS

Languages	Python (Proficient), Java, SQL
AI / ML Technologies	PyTorch, TensorFlow, scikit-learn, Hugging Face, LangChain, OpenCV
Developer Tools	Git, Docker, AWS, Azure, MLflow, CI/CD, RESTful APIs
Core Competencies	Machine Learning, Generative AI (LLMs/RAG), NLP, MLOps, API Design

ACHIEVEMENTS & ACTIVITIES

- Competitive Programming:** Solved 300+ problems on **LeetCode**, demonstrating strong proficiency in data structures, algorithms, and complex problem-solving.
- Machine Learning Specialization:** Completed the **DeepLearning.AI certification** focusing on advanced neural networks and ML architectures.